Name: _____

TOPIC TEST 1

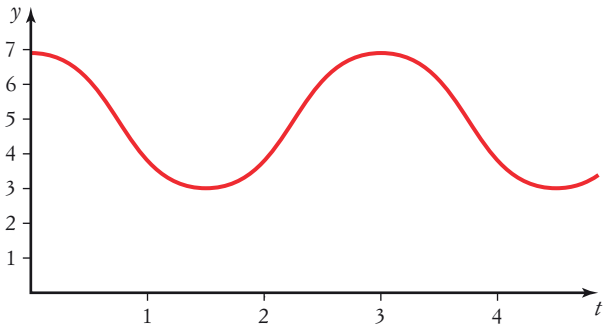
Chapter 6: Trigonometric functions

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks)
- Part B: 7 short answer questions (38 marks)
- Part C: 2 analysis questions (11 marks)

Part A

5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 Find the amplitude and period of the following graph.



A Amplitude 4, period 4

B Amplitude 4, period 3

C Amplitude 4, period $\frac{2\pi}{3}$

D Amplitude 2, period 3

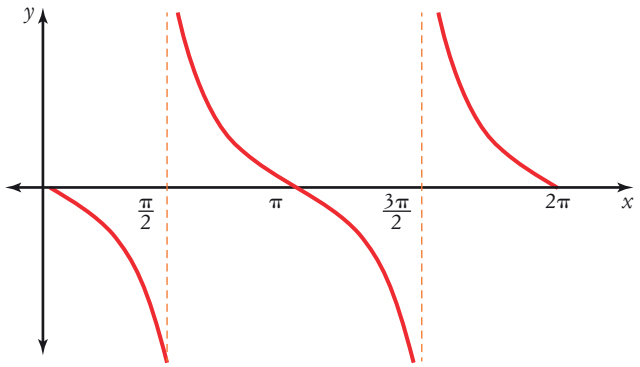
E Amplitude 2, period $\frac{2\pi}{3}$

[1 mark]

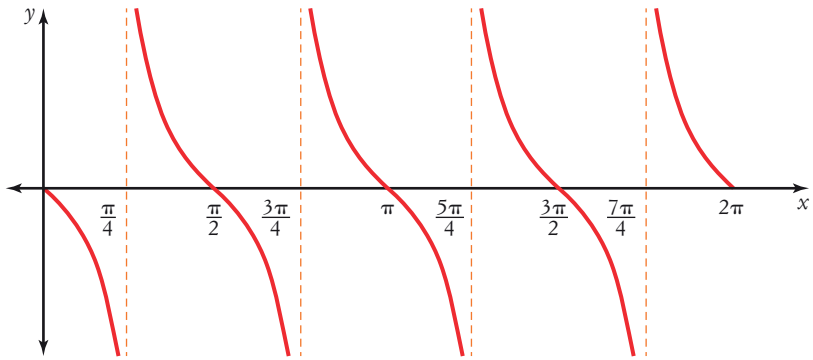
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2 The graph of $y = \tan(2x)$ is most like:

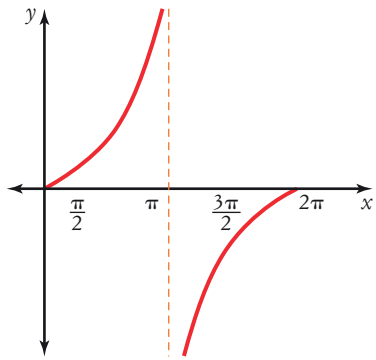
A



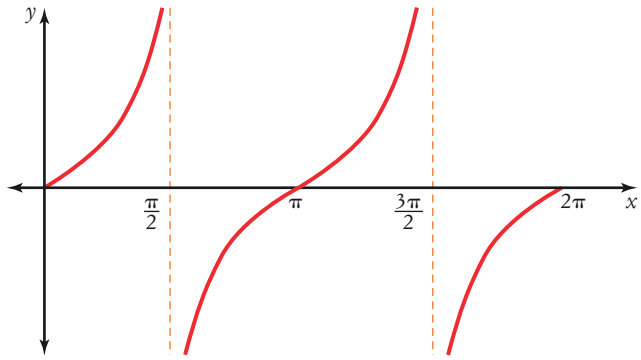
B



C

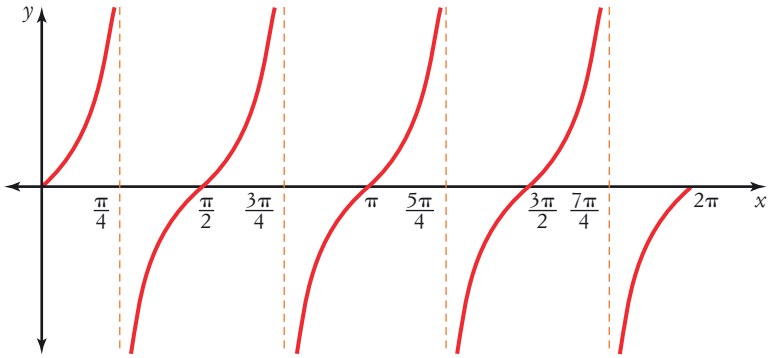


D



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E



[1 mark]

3 Which statement is not correct?

A $\sin\left(x + \frac{\pi}{2}\right) = \cos(x)$

B $\cos\left(\frac{\pi}{2} + x\right) = \sin(x)$

C $\cos\left(x - \frac{\pi}{2}\right) = \sin(x)$

D $\sin\left(\frac{\pi}{2} - x\right) = \cos(x)$

E $\cos\left(\frac{\pi}{2} - x\right) = \sin(x)$

[1 mark]

4 Which statement is not correct?

A $\cos(x + y) = \cos(x)\cos(y) - \sin(x)\sin(y)$

B $\sin(x + y) = \sin(x)\cos(y) + \cos(x)\sin(y)$

C $\tan(x - y) = \dfrac{\tan(x) - \tan(y)}{1 + \tan(x)\tan(y)}$

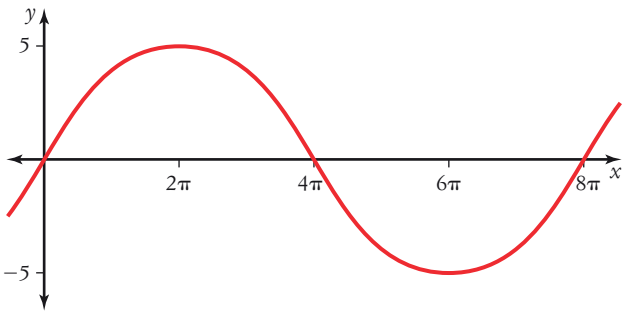
D $\sin(x - y) = \sin(y)\cos(x) - \cos(y)\sin(x)$

E $\tan(x + y) = \dfrac{\tan(x) + \tan(y)}{1 - \tan(x)\tan(y)}$

[1 mark]

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5 The equation of the function below is



A $y = 8 \sin(5x)$

B $y = 5 \sin(8x)$

C $y = 5 \sin\left(\frac{\pi x}{8}\right)$

D $y = 5 \sin\left(\frac{\pi x}{4}\right)$

E $y = 5 \sin(4\pi x)$

[1 mark]

Part B

7 short answer questions
38 marks
Show your working where appropriate.

1 Find the value of

a $\tan(360°)$ _____

b $\sin\left(\frac{\pi}{2}\right)$ _____

c $\cos(180°)$ _____

d $\sin(2\pi)$ _____

e $\tan(0°)$ _____

[5 marks]

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2 For the domain $0 \leq x \leq 2\pi$, sketch the graph of:

a $y = \cos(x)$

b $y = 2 - \sin(x)$

c $y = 2\sin(x) - 1$

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d $y = 6\cos\left(\frac{x}{2}\right)$

e $y = \tan\left(x - \frac{\pi}{4}\right)$

[10 marks]

3 If $\sin(a) = \frac{4}{7}$ and $\cos(b) = \frac{3}{5}$, find the exact value of $\cos(a + b)$

[5 marks]

4 By drawing the graphs of $y = 2\cos(x)$ and $y = x - 1$ on the same set of axes, graphically solve $2\cos(x) = x - 1$. You may use a CAS calculator.

[4 marks]

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5 Show that $\cos(2x) = 1 - 2\sin^2(x)$

[3 marks]

6 Solve for $0 \leq x \leq 2\pi$

a $\sin(x) = \frac{1}{2}$

b $\cos(x) = -\frac{1}{2}$

[6 mark]

7 Solve $2\sin^2(\theta) - 1 = 0$ for $0° \leq \theta \leq 360°$.

[5 marks]

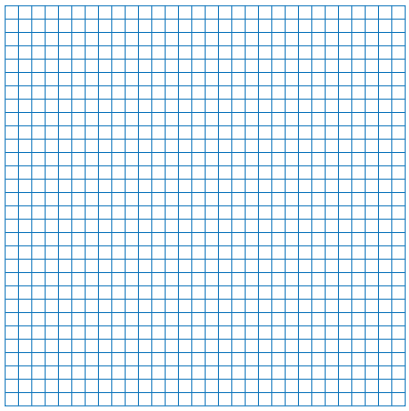
Part C

2 analysis questions
11 marks
Show your working where appropriate.

1 Below is a table showing the average number of daylight hours over several months.

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug
Daylight hours	15.3	14.7	13.2	13.1	12.7	12.2	12.5	13.8

a Draw a graph to display this data.



b Is it periodic? If so, what is the period?

c Find the amplitude.

d Approximately how many daylight hours would you expect there to be in November?

[6 marks]

2 The sales, S, in 100s of units, of a seasonal product are modelled by

$S = 54.8 + 32.5 \cos\left(\frac{\pi t}{6}\right)$

where t is the time in months ($t = 1$ is January and $t = 12$ is December).

a Sketch a graph of the sales for a period of 12 months.

b Use the graph to find the months when sales were greater than 6800 units.

[5 marks]

Total marks: 54

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Answers

Part A

[1 mark each]

1 Amplitude $= \dfrac{7-3}{2} = 2$

Period $= 3$

$\therefore$ **D** [1 mark]

2 Period $= \dfrac{\pi}{b} = \dfrac{\pi}{2}$

$\therefore$ **E** [1 mark]

3 $\sin\left(x + \dfrac{\pi}{2}\right) = \cos(x)$ and $\cos\left(x - \dfrac{\pi}{2}\right) = \sin(x)$ (phase changes)

$\sin\left(\dfrac{\pi}{2} - x\right) = \cos(x)$ and $\cos\left(\dfrac{\pi}{2} - x\right) = \sin(x)$ (from right-angled triangle)

$\therefore$ **B** [1 mark]

4 $\sin(x - y) = \sin(x)\cos(y) - \cos(x)\sin(y)$

$\therefore$ **D** [1 mark]

5 Amplitude $= 5$

Period $= 8$

$\dfrac{2\pi}{b} = 8$

$\dfrac{2\pi}{8} = b$

$\dfrac{\pi}{4} = 8$

$y = a\sin(bx)$

$= 5\sin\left(\dfrac{\pi x}{4}\right)$

$\therefore$ **D** [1 mark]

Part B

1 **a** $\tan(360°) = 0$ [1 mark]

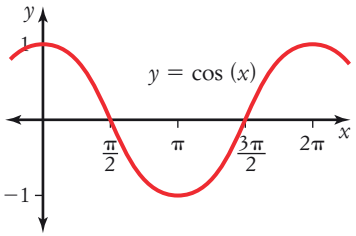
b $\sin\left(\dfrac{\pi}{2}\right) = 1$ [1 mark]

c $\cos(180°) = -1$ [1 mark]

d $\sin(2\pi) = 0$ [1 mark]

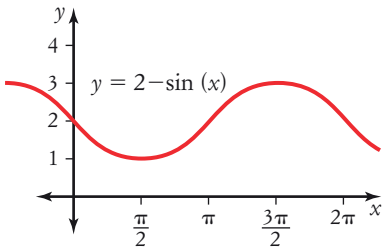
e $\tan(0°) = 0$ [1 mark]

2 a Period $= 2\pi$, amplitude 1 **[1 mark]**



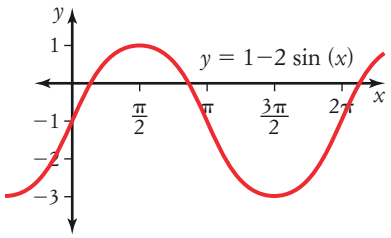
[1 mark]

b Period $= 2\pi$, amplitude 1, $-\sin(x)$ is reflected in the x-axis, translated 2 units up. **[1 mark]**



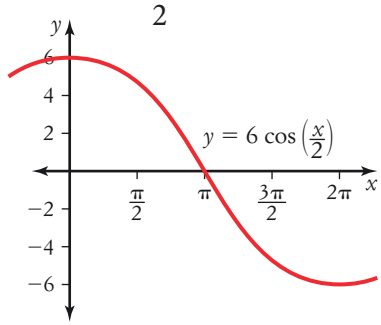
[1 mark]

c Period $= 2\pi$, amplitude $= 2$, translated 1 unit down. **[1 mark]**



[1 mark]

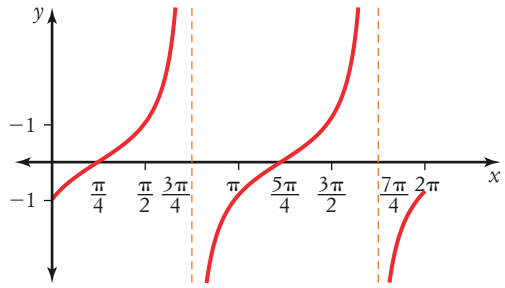
d Period $= \dfrac{2\pi}{\frac{1}{2}} = 4\pi$, amplitude $= 6$ **[1 mark]**



[1 mark]

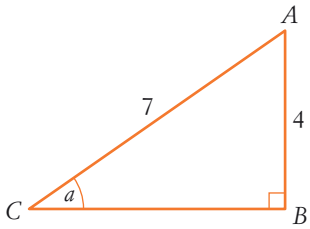
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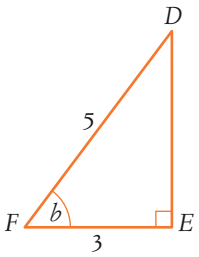
e The graph $y = \tan(x)$ is translated $\frac{\pi}{4}$ units to the right. **[1 mark]**



[1 mark]

3





$CB^2 = 7^2 - 4^2 = 33$

$CB = \sqrt{33}$ **[1 mark]**

$DE = 4$ (3-4-5 triangle) **[1 mark]**

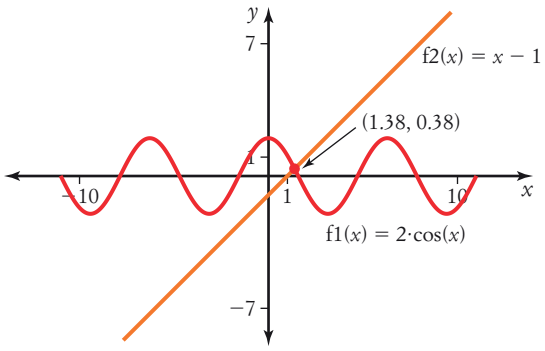
$\cos(a + b) = \cos(a)\cos(b) - \sin(a)\sin(b)$ **[1 mark]**

$$= \frac{\sqrt{33}}{7} \times \frac{3}{5} - \frac{4}{7} \times \frac{4}{5}$$ **[1 mark]**

$$= \frac{3\sqrt{33}}{35} - \frac{16}{35}$$

$$= \frac{3\sqrt{33} - 16}{35}$$ **[1 mark]**

4 On the CAS calculator or freehand, draw both graphs.



[3 marks]

$x = 1.4$ **[1 mark]**

5 $\cos(2x) = \cos(x + x)$

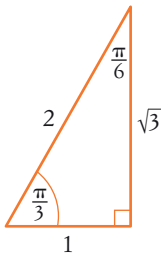
$= \cos(x)\cos(x) - \sin(x)\sin(x)$ **[1 mark]**

$= \cos^2(x) - \sin^2(x)$ **[1 mark]**

$= 1 - \sin^2(x) - \sin^2(x)$

$= 1 - 2\sin^2(x)$ **[1 mark]**

6 a From the special triangle, $\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$ **[1 mark]**



$x = \frac{\pi}{6}, \pi - \frac{\pi}{6}$ (sin is positive in 1st, 2nd quadrants) **[1 mark]**

$= \frac{\pi}{6}, \frac{5\pi}{6}$ **[1 mark]**

b From the triangle, $\cos\left(\frac{\pi}{3}\right) = \frac{1}{2}$ **[1 mark]**

$x = \pi - \frac{\pi}{3}, \pi + \frac{\pi}{3}$ (cos is negative in 2nd, 3rd quadrants) **[1 mark]**

$= \frac{2\pi}{3}, \frac{4\pi}{3}$ **[1 mark]**

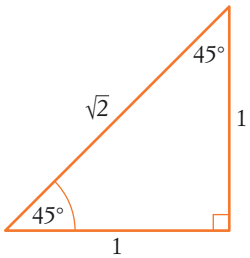
7 $2\sin^2(\theta) = 1$

$\sin^2(\theta) = \frac{1}{2}$ **[1 mark]**

$\sin(\theta) = \pm\sqrt{\frac{1}{2}}$

$= \pm\frac{1}{\sqrt{2}}$ **[1 mark]**

From the special triangle, $\sin(45°) = \frac{1}{\sqrt{2}}$

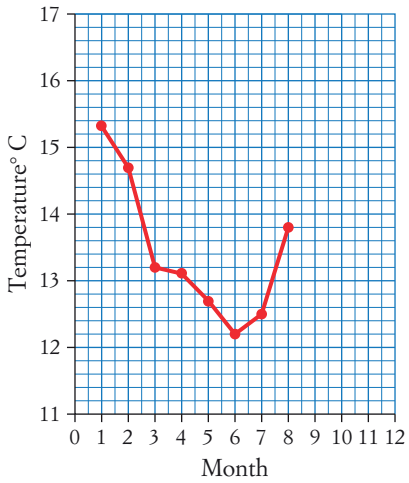


$\theta = 45°, 180° - 45°, 180° + 45°, 360° - 45°$ (all quadrants) **[2 marks]**

$= 45°, 135°, 225°, 315°$ **[1 mark]**

Part C

1 a



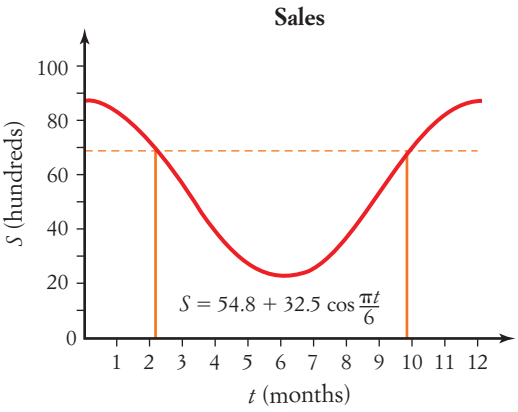
[3 marks]

b It is hard to tell if this graph is periodic without more data, but common sense says it would be 12 months. **[1 mark]**

c Amplitude $= \dfrac{15.3 - 12.2}{2} \approx 1.6$ **[1 mark]**

d In November you would expect the number of daylight hours to increase, say to around 15 hours, from extrapolating the graph. **[1 mark]**

2 a Period $= \dfrac{2\pi}{\frac{\pi}{6}} = 2\pi \times \dfrac{6}{\pi} = 12$; amplitude $= 32.5$; the graph is translated 54.8 units upwards. **[1 mark]**



[2 marks]

b Draw a horizontal line across the graph at 6800.

The months that exceed 6800 units are Jan, Feb, Oct, Nov and Dec. **[2 marks]**

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